Empirical Methods for Policy Evaluation

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Motivation/Background

- Debate on economic theory $\Leftrightarrow$ econometric modeling and estimation
  - $\Rightarrow$ Design-based vs. model-based approaches to policy evaluation
- Natural synergy for a better characterization of policy impacts
  - $\Rightarrow$ Middle ground approach dates back to Marschak (1953)
  - $\Rightarrow$ Many advocates since then....
- We will focus on both methods and applications (Labor, Devo, Public,..)

The Causal Inference (aka Design-Based) Approach

- Binary random variable $D_i = \{0, 1\}$ and potential outcomes (Y_i^1, Y_i^0)

$$\begin{aligned} \text{ATE} &= \mathbb{E}(Y_i^1 - Y_i^0) \\ &= \mathbb{E}(Y_i^1 \mid D_i = 1) - \mathbb{E}(Y_i^0 \mid D_i = 0) \\ &= \underbrace{\mathbb{E}(Y_i^1 - Y_i^0 \mid D_i = 1)}_{\text{ATT}} + \underbrace{\mathbb{E}(Y_i^0 \mid D_i = 1) - \mathbb{E}(Y_i^0 \mid D_i = 0)}_{\text{Selection bias}} \end{aligned}$$

- Research designs attempt to eliminate bias by means of assumptions. E.g.:
  - ⇒ Continuity of potential outcomes around a cutoff
  - ⇒ Same trend in potential outcomes over time

The Structural (aka model-based) Approach

- Lay out an economic model of the phenomenon being studied
- Addition of a stochastic structure if the model itself does not possess one
- Identification of the parameters given the data and model
  - $\Rightarrow$ Two sets of parameters yielding the same likelihood are necessarily equal
  - $\Rightarrow$ Observables pick only one set of parameters irrespectively of unobservables
- Estimation technique given model, identification, and data
- Empirical comparative statics, welfare metrics, and/or policy experiments

Course Outline and Required Readings

0. Causal inference $\Leftrightarrow$ Structural models
   - $\Rightarrow$ The effect of cash grants on school attainment (Attanasio-Meghir-Santiago, 2012 Restud)
1. Randomized experiments $\Leftrightarrow$ Dynamic latent factor models
   - $\Rightarrow$ Production of human capital in early years (Attanasio-Cattan-Fitzsimons-Meghir-Rubio, 2020 AER)
2. Regression discontinuity designs $\Leftrightarrow$ School choice models
   - $\Rightarrow$ Teacher sorting and student achievement in Peru (Bobba-Ederer-Leon-Neilson-Nieddu, Forth. JPE)
3. Difference-in-difference and event study designs $\Leftrightarrow$ Job search models
   - $\Rightarrow$ Informal labor and schooling investments in Mexico (Bobba-Flabbi-Levy, 2022 IER)
4. Instrumental variables and shift-share designs $\Leftrightarrow$ Models of firm dynamics
   - $\Rightarrow$ Rural-urban migration in Brazil (Imbert-Ulyssea, 2026 ECMA)

Course Material

- Slides are the main reference for the course, also on `matteobobba.github.io/teaching/EMPE`
- Survey papers of causal inference methods discussed in class
  - ⇒ Athey-Imbens (2017, HFE), "The Econometrics of Randomized Experiments"
  - ⇒ Cattaneo-Titiunik (2022, ARE), "Regression Discontinuity Designs"
  - ⇒ De Chaisemartin-D'Haultfœuille (2023, EctJ), "Two-Way Fixed Effects and Difference-in-Differences with Heterogeneous Treatment Effects: A Survey"
  - ⇒ Boryusak-Hull-Jaravel (2025, JEP), "A Practical Guide to Shift-Share Instruments"

Course Material (continued)

- Survey papers of structural methods discussed in class
  - ⇒ Keane-Todd-Wolpin (2011, HoLE), "The Structural Estimation of Behavioral Models: Discrete Choice Dynamic Programming Methods and Applications"
  - ⇒ Cunha-Nielsen-Williams (2021, ARE), "The Econometrics of Early Childhood Human Capital and Investments"
  - ⇒ Agarwal-Somaini (2020, ARE), "Revealed Preference Analysis of School Choice Models"
  - ⇒ Eckstein-van den Berg (2007, JoE), "Empirical Labor Search: A Survey "

Course Material (continued)

- Survey papers of mixed approaches discussed in class
  - ⇒ Todd-Wolpin (2021, JEP), "The Best of Both Words: Combining RCTs with Structural Modeling"
  - ⇒ Mahoney (2022, JEP), "Principles for Combining Descriptive and Model-Based Analysis in Applied Microeconomics Research"
  - ⇒ Galiani-Pantano (2023, Handbook Ch.), "Structural Models: Inception and Frontier"
  - ⇒ Low-Meghir (2017, JEP), "The use of structural models in econometrics"

Course Evaluation

1. Three problem sets [3×25% of the grade]
   - ⇒ Coding-intensive exercises based on papers discussed in class
   - ⇒ You can work in (different) pairs across assignments
   - ⇒ Due on weeks 3, 6, and 9
2. One referee report [25% of the grade]
   - ⇒ Pick one paper from a list that I will circulate around weeks 5-6
   - ⇒ Individual assignment
   - ⇒ Due by the end of semester (mid January)

Causal Inference Meets Structural Models (and Viceversa)

1. Ex-ante policy evaluation

   ⇒ Chapter 2 in Wolpin's book (MIT press, 2013)

2. Dynamic discrete choice models

   ⇒ Keane-Todd-Wolpin (2011, HoLE)

3. The effect of cash grants on schooling attainment

   ⇒ Todd-Wolpin (AER, 2006)

   ⇒ Attanasio-Meghir-Santiago (Restud, 2012)

4. Principles for combining design-based and model-based analysis

Ex-ante policy evaluation

The Structural Approach for Evaluating Public Policies

- Economic models allow predicting the effects of public policies
  - $\Rightarrow$ Before they are implemented and/or variants of existing policies
- Model-based approach is particularly suitable for informing policy
  - $\Rightarrow$ Improve program design so as to maximize impacts/welfare given costs
  - $\Rightarrow$ Inform program targeting by identifying sub-populations for which impacts/welfare are highest
  - $\Rightarrow$ Study the indirect effect of programs beyond beneficiaries due to spillover and/or GE effects
  - $\Rightarrow$ Analyze program impacts over a different time horizon than the actual variation in the data

An Example

- Many governments have adopted conditional cash transfer (CCT) programs
  - ⇒ Provide cash transfers to HHs conditional on school attendance of children
  - ⇒ Alleviate poverty and stimulate human capital investments
- Can we evaluate those programs before they are implemented?
  - ⇒ A model of schooling decisions where transfer decreases schooling costs

Economic Model

- Consider the following (static) optimization problem for the household

$$\max_{s \in \{0,1\}} U(c, s) \text{ s.t. } \begin{cases} c = y + w(1-s) \\ c = y + w(1-s) + \tau s \end{cases}$$

- Optimal schooling choices without and with the subsidy

$$s^\star = g(y, w)$$
$$s^{\star\star} = g(\tilde{y}, \tilde{w}), \text{ where } \tilde{y} = y + \tau \text{ and } \tilde{w} = w - \tau$$

- The subsidy acts as an income-compensated reduction in child wages

Identification

- Add observable (X) and unobservable (ϵ) preference shifters

$$U(c, s; X, \epsilon)$$

- Unobserved heterogeneity is not systematically related to wages and income

$$f(\epsilon|y, w, X) = f(\epsilon|X) \tag{CIA}$$

- Given CIA, variations in wages and income identify the impact of the program

Estimation and Counterfactuals

- Ex-ante average treatment effect is:

$$\text{ExATE}_{np} = \frac{1}{N}\sum_{j=1}^{N}\left[\underbrace{\hat{\mathbb{E}}(s_i \mid w_i = w_j - \tau_j, y_i = y_j + \tau_j, X_i)}_{\text{Predicted schooling under the program}} - \underbrace{s_j(w_j, y_j, X_j)}_{\text{Observed schooling}}\right]$$

- $\mathbb{E}(s_i \mid \tilde{w}_i, \tilde{y}_i, X_i)$ can be estimated non-parametrically (Kernel, LLR, or series)
  - $\Rightarrow$ Need common support in the data: $(w, y) \in R^2$ such that $f(w, y) > 0$

Change in the Grant Amount

	Boys		
Ages	2* Original	Original	0.75*Original
12-13	0.04 (59%)	0.01 (87%)	0.003 (98%)
14-15	0.24 (45%)	0.01 (83%)	0.05 (98%)
12-15	0.12 (53%)	0.06 (86%)	0.02 (98%)
	Girls		
	2* Original	Original	0.75*Original
12-13	0.06 (48%)	0.06 (91%)	0.05 (98%)
14-15	0.23 (51%)	0.07 (89%)	0.03 (98%)
12-15	0.14 (50%)	0.06 (90%)	0.05 (98%)
	Boys and Girls		
	2* Original	Original	0.75*Original
12-13	0.05 (54%)	0.04* (89%)	0.03 (98%)
14-15	0.23 (48%)	0.09 (86%)	0.04 (98%)
12-15	0.13 (52%)	0.06 (88%)	0.03 (98%)

† Bandwidth equals 200 pesos. Trimming implemented using the 2% quantile of positive density values as the cut-off point.

Unconditional Income Grant

	Boys		
Ages	Predicted	Sample-Sizes‡	% overlapping support
12-13	-0.02 (0.03)	374, 610	89%
14-15	-0.06 (0.05)	309, 569	90%
12-15	-0.04 (0.03)	683, 1179	89%
	Girls		
	Predicted	Sample-Sizes‡	% overlapping support
12-13	-0.03 (0.04)	361, 589	88%
14-15	0.00 (0.05)	316, 591	88%
12-15	-0.02 (0.03)	677, 1180	88%
	Boys and Girls		
	Predicted	Sample-Sizes‡	% overlapping support
12-13	-0.03 (0.03)	735, 1199	88%
14-15	-0.03 (0.03)	625, 1160	89%
12-15	-0.03 (0.02)	1360, 2359	89%

†Standard errors based on 500 bootstrap replications. Bandwidth equals 200 pesos. Trimming implemented using the 2% quantile of positive density values as the cut-off point.

Model Extension: Adding Home Production

- Allow for an alternative use of children's time, home production $l \in \{0, 1\}$

$$\max_{(s,l)} U(c, l, s) \text{ s.t. } \begin{cases} c = y + w(1 - s - l) \\ c = y + w(1 - s - l) + \tau s \end{cases}$$

- Optimal schooling choices without and with the subsidy are different

$$s^\star = g(y, w)$$
$$s^{\star\star} = h(\tilde{y}, \tilde{w}, \tau)$$

- Non-parametric ex-ante approach is not feasible in this case

Estimation

- Consider the following functional form (no child leisure and no X)

$$U(C, s; \epsilon) = C + \alpha s + \beta C s + \epsilon s, \ \epsilon \sim N(0, \sigma_\epsilon^2)$$

- The probability of school attendance under the subsidy τ is

$$P(s = 1) = 1 - \Phi\left(\frac{(w - \tau) - \alpha - \beta(y + \tau)}{\sigma_\epsilon}\right)$$

- Model parameters can be estimated by ML from data with $\tau = 0$ given variation in (w, y)

Counterfactual

- Given parameter estimates, the effect of the subsidy τ on the attendance rate is

$$\text{ExATE}_p = \Phi\left(\frac{(w-\tau)-\hat{\alpha}-\hat{\beta}(y+\tau)}{\hat{\sigma}_\epsilon}\right) - \Phi\left(\frac{w-\hat{\alpha}-\hat{\beta}y}{\hat{\sigma}_\epsilon}\right)$$

- There is no condition on the support of y and w

Wrapping Up on Ex-Ante Policy Evaluation

- Estimating ex-ante policy impact does not require specifying a full model
- Given model restrictions, nonparametric approaches may be feasible
  - $\Rightarrow$ Even when there is no variation in the data on the policy (price of schooling)
- If not feasible, need to impose distributional assumptions on model primitives

Dynamic discrete choice models

The Importance of Dynamics to Evaluate a CCT Program

- Child's wage may increase with past work experience
- Past education could change attitudes towards attendance
- Parents' utility=f(stock of educ.), so current attendance affects future utility
- The incentives of the grant itself creates dynamics
  - $\Rightarrow$ Not going to school one year reduces the number of years of the subsidy
  - $\Rightarrow$ The grant is only available until age 17

Decision Problem

- At each discrete period $t = 1, ..., T$, an individual chooses among J mutually exclusive alternatives

$$d_{jt} = \begin{cases} 1 & \text{if } j \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

- Individual utility in period t depends on the vector of state variables $\boldsymbol{s_t} = (\boldsymbol{x_t}, \boldsymbol{\epsilon}_t)$

  $\Rightarrow$ State variables follow a choice-specific Markovian process: $\boldsymbol{s_{t+1}} \sim F(\boldsymbol{s_{t+1}} | \boldsymbol{s_t}, d_{t+1})$

- Optimal decision at time t:

$$d^{\star}_{jt} = \arg\max_{d_t \in \mathcal{D}} \mathbb{E}_t \left[\sum_{l=0}^{T-t} \beta_l U(\boldsymbol{s_t}, d_{t+1}) \right]$$

Baseline Assumptions

1. Additive separability: $U(d_t, \boldsymbol{x_t}, \boldsymbol{\epsilon}_t) = u(d_t, \boldsymbol{x_t}) + \epsilon_t(d)$

   $\Rightarrow$ $u(d_t, \boldsymbol{x_t}) \equiv \sum_j d_{jt} u_{jt}(\boldsymbol{x_t})$ and $\epsilon_t(d) \equiv \sum_j d_{jt}\epsilon_{jt}$

2. ϵ_t are iid across agents and over time $\sim F_\epsilon(\boldsymbol{\epsilon}_t)$

3. Conditional independence of future x: $F_x(\boldsymbol{x_{t+1}}|d_t, \boldsymbol{x_t}, \boldsymbol{\epsilon}_t) = F_x(\boldsymbol{x_{t+1}}|d_t, \boldsymbol{x_t})$

   $\Rightarrow$ Assumptions (2) and (3) imply $F(x_{t+1}, \epsilon_{\mathbf{t+1}}|d_t, \mathbf{x_t}, \epsilon_{\mathbf{t}}) = F_x(x_{t+1}|d_t, \mathbf{x_t}, \epsilon_{\mathbf{t}})F_\epsilon(\epsilon_{\mathbf{t+1}})$

4. Logit: ϵ_{jt} are iid across alternatives and type-I extreme value distributed

Value Functions and Bellman Principle

- Let $V_t(\boldsymbol{x_t})$ denote the ex-ante value function in period t

$$V_t(\boldsymbol{x_t}) = \mathbb{E}_{t-1}\left[\sum_{l=0}^{T-t}\sum_{j=1}^{J}\beta^l d^{\star}_{jt+l}(u_{jt+l}(\boldsymbol{x}_{t+l}) + \epsilon_{jt+l}) \mid \boldsymbol{x_t}\right] \tag{Emax}$$

- Using the Bellman's optimality principle we can write:

$$\begin{aligned} V_t(\boldsymbol{x_t}) &= \mathbb{E}_{t-1}\left[\sum_{j}^{J} d^{\star}_{jt}\left(u_{jt}(\boldsymbol{x}_t) + \epsilon_{jt} + \beta\int V_{t+1}(\boldsymbol{x}_{t+1})dF_x(\boldsymbol{x}_{t+1} \mid \boldsymbol{x}_t, d^{\star}_t)\right) \mid \boldsymbol{x_t}\right] \\ &= \sum_{j=1}^{J}\int d^{\star}_{jt}\left(u_{jt}(\boldsymbol{x}_t) + \epsilon_{jt} + \beta\int V_{t+1}(\boldsymbol{x}_{t+1})dF_x(\boldsymbol{x}_{t+1} \mid \boldsymbol{x}_t, d^{\star}_t)\right) dF_\epsilon(\boldsymbol{\epsilon}_t) \end{aligned}$$

Conditional Choice Probabilities (CCPs)

- Define the conditional value function $v_{jt}(\boldsymbol{x_t})$ as the deterministic payoff of option j:

$$v_{jt}(\boldsymbol{x_t}) \equiv u_{jt}(\boldsymbol{x}_t) + \beta \int V_{t+1}(\boldsymbol{x}_{t+1}) dF_x(\boldsymbol{x}_{t+1} \mid \boldsymbol{x}_t, j)$$

- Then, the individual chooses j in period t iff

$$v_{jt}(\boldsymbol{x_t}) + \epsilon_{jt} \geq v_{kt}(\boldsymbol{x_t}) + \epsilon_{kt} \;\; \forall k \neq j$$

- Given Logit errors, CCPs are

$$p_{jt}(\boldsymbol{x_t}) \equiv \mathbb{E}[d^{\star}_{jt} \mid \boldsymbol{x}_t] = \frac{e^{v_{jt}(\boldsymbol{x_t})}}{\sum_h e^{v_{ht}(\boldsymbol{x_t})}}$$

Conditional Choice Probabilities (continued)

- Emax with Logit errors simplifies to

$$V_t(\boldsymbol{x_t}) = \ln \sum_{j=1}^{J} \exp\{v_{jt}(\boldsymbol{x_t})\} + \gamma$$

- In that case the conditional value function becomes:

$$v_{jt}(\boldsymbol{x_t}) = u_{jt}(\boldsymbol{x}_t) + \beta \int \ln \sum_{j=1}^{J} \exp\{v_{jt+1}(\boldsymbol{x}_{t+1})\} dF_x(\boldsymbol{x}_{t+1} \mid \boldsymbol{x}_t, j) + \beta\gamma$$

- Find $\{v_{jt}(\boldsymbol{x_t})\}_{j=1,\ldots,J}^{t=1,\ldots,T}$ with backward induction (finite horizon) or fixed point (infinite horizon)

Likelihood Function

- We have longitudinal data for a sample $\{d_{it}, \boldsymbol{x}_{it}\}_{i=1,\ldots,N}^{t=1,\ldots,T_i}$
- The log-likelihood of this sample is given by:

$$L_N(\theta) = \sum_{i=1}^{N} \ln Pr(d_{i1}, d_{i2}, \ldots, d_{iT_i}, \boldsymbol{x}_{i1}, \boldsymbol{x}_{i2}, \boldsymbol{x}_{iT_i}; \boldsymbol{\theta}) \equiv \sum_{i=1}^{N} l_i(\boldsymbol{\theta})$$

- Given conditional independence, this probability can be factorized as:

$$l_i(\boldsymbol{\theta}) = \underbrace{\sum_{t=1}^{T_i} \ln Pr(d_{it} \mid \boldsymbol{x_t}; \boldsymbol{\theta})}_{\text{Sum of CCPs}} + \underbrace{\sum_{t=2}^{T_i} \ln Pr(\boldsymbol{x}_{it} \mid \boldsymbol{x}_{it-1}, d_{it-1}; \boldsymbol{\theta})}_{F_x(\boldsymbol{x}_{it} \mid \boldsymbol{x}_{it-1}, d_{it-1})} + \underbrace{\ln Pr(\boldsymbol{x}_{i1}; \boldsymbol{\theta})}_{\text{Initial condition}}$$

Unobserved Heterogeneity

- Assuming that unobservable characteristics are not correlated over time might be restrictive
- Let ω_i be individual-specific time-invariant component with support $\Omega \equiv \{\omega_i^k : k = 1, 2, .., K\}$
  - $\Rightarrow$ Introduces persistence in unobservable state variables
  - $\Rightarrow$ $\boldsymbol{x_{it}}$ is no longer a sufficient statistic for d_{it} (lagged decisions $d_{it-1}, ..., d_{i1}$ contain info about $\boldsymbol{\omega}$)
- Unconditional log-likelihood is

$$L_N(\theta) = \sum_{k=1}^{K}\sum_{i=1}^{N} \ln P(d_{i1}, d_{i2}, ..., d_{iT_i}, \boldsymbol{x}_{i1}, \boldsymbol{x}_{i2}, \boldsymbol{x}_{iT_i} | \omega_i^k; \boldsymbol{\theta}) P(\omega_i = \omega_i^k | \boldsymbol{x}_{i1})$$

  - $\Rightarrow$ Likelihood factorization not feasible since conditional independence breaks with ω_i
  - $\Rightarrow$ Initial conditions $\boldsymbol{x}_{i1}$ are assumed exogenous conditional on type ω

The effect of cash grants on schooling attainment

The *Progresa* Program in Mexico

- Large scale anti-poverty program
  - Began in 1997 in rural areas and rapidly expanded throughout the country
  - About 20% of Mexican families participating
- Educational grants to mothers to encourage children's school attendance
  - Benefits levels increase with grades attained, higher for girls
  - Subsidies amount to about 20% of average annual income
- Data from the initial rural evaluation of the program

  $\Rightarrow$ Randomized phase-in design at the village level
  - Within villages, both eligible and non-eligible HHs (wealth index)

Todd and Wolpin (AER 2006)

- Dynamic discrete choice of children's time allocation and family fertility
  - $\Rightarrow$ Ex-ante evaluation (control group): MU of subsidy=MU of income
- Each year a married couple decides on whether
  - $\Rightarrow$ Each of their children attend school/stay-at-home/work for wage
  - $\Rightarrow$ The wife becomes pregnant
- Parental and children earnings are subject to idiosyncratic time-varying shocks

Model Validation: Within-Sample Fit (Control Group)

Boys							
	Actual			Predicted			
Age	School	Work	Home	School	Work	Home	χ^2
6	0.933	—	0.066	0.923	—	0.077	0.58
7	0.981	—	0.019	0.980	—	0.020	0.02
8	0.987	—	0.013	0.980	—	0.020	0.99
9	0.994	—	0.006	0.979	—	0.021	3.49
10	0.982	—	0.018	0.974	—	0.026	0.86
11	0.977	—	0.023	0.964	—	0.036	1.45
12	0.885	0.021	0.094	0.846	0.039	0.115	3.99
13	0.780	0.084	0.136	0.736	0.078	0.186	4.51
14	0.677	0.157	0.166	0.619	0.191	0.190	3.41
15	0.490	0.276	0.235	0.520	0.251	0.229	0.88
Girls							
6	0.965	—	0.035	0.942	—	0.058	3.84
7	0.976	—	0.024	0.968	—	0.032	0.77
8	0.989	—	0.011	0.976	—	0.024	1.96
9	0.991	—	0.009	0.975	—	0.025	3.26
10	0.979	—	0.021	0.970	—	0.030	0.93
11	0.969	—	0.031	0.948	—	0.052	2.97
12	0.896	0.007	0.097	0.854	0.020	0.126	4.61
13	0.726	0.028	0.245	0.676	0.025	0.299	2.85
14	0.582	0.089	0.329	0.566	0.092	0.342	0.22
15	0.419	0.123	0.458	0.402	0.157	0.442	1.68

Note: χ^2 (0.05, 1) = 3.84, χ^2 (0.05, 2) = 5.99.

Design-based $\Rightarrow$ Model-based: Out of Sample Validation

- The (im)plausibility of the model assumptions does not determine the credibility of its predictions
  - $\Rightarrow$ Within-sample goodness-of-fit tests are necessary but not sufficient
- The credibility of counterfactual predictions is better assessed in terms of out-of-sample fit
  - $\Rightarrow$ Estimate a model by holding out the treatment/control group
  - $\Rightarrow$ Validate its predictions about program impacts through model-based counterfactuals

Out-of-Sample Model Validation Using the Experiment

	Girls age 12–15			Girls age 12–15, behind in school			Girls age 13–15, HGC ≥ 6, behind in school		
	(1) Actual	(2) Pred. with Subsidy	(2)–(1)	(1) Actual	(2) Pred. with Subsidy	(2)–(1)	(1) Actual	(2) Pred. with Subsidy	(2)–(1)
97 Control	65.3	72.7	7.4	58.3	67.0	8.7	40.9	58.6	17.7
98 Control	66.5	72.9	6.4	58.7	66.9	8.2	44.4	60.6	16.2
97 Treatment	62.9	73.0	10.1	56.9	67.6	10.7	30.3	56.2	25.9
Experimental treatment effect:									
Cross section		8.0 (4.6)			12.8 (5.7)			7.1 (8.6)	
Difference-in-difference		10.3 (6.7)			14.1 (8.3)			17.7 (12.0)	

	Boys age 12–15			Boys age 12–15, behind in school			Boys age 13–15, HGC ≥ 6, behind in school		
	(1)	(2)	(2)–(1)	(1)	(2)	(2)–(1)	(1)	(2)	(2)–(1)
97 Control	68.8	79.6	10.8	64.0	75.8	11.8	59.0	72.7	13.7
98 Control	72.5	80.2	7.7	67.4	78.0	10.6	57.1	72.8	15.7
97 Treatment	69.5	79.4	9.9	64.2	75.8	11.6	52.6	71.6	19.0
Experimental treatment effect:									
Cross section		3.8 (4.2)			4.2 (5.2)			1.2 (8.4)	
Difference-in-difference		3.1 (6.1)			4.0 (7.4)			3.8 (11.7)	

Design-based $\Rightarrow$ Model-based: Identification/Estimation

- Use policy variation for estimation of the model parameters
  - $\Rightarrow$ Allows researcher to relax some behavioral/distributional assumptions
- In the *Progresa* case, MU of subsidy $\neq$ MU of income
  - $\Rightarrow$ Transfers to mothers, and who receives the money likely matters

Attanasio, Meghir and Santiago (ReStud, 2012)

- Similar dynamic discrete choice structure with some differences
  - ⇒ No fertility decision
  - ⇒ Binary choice: school vs. work
  - ⇒ Each child's utility is independent of that of the parents/other children
  - ⇒ Allow for MU of the subsidy to differ from MU of other sources of income
  - ⇒ Allow for equilibrium effects of the program on children's wages

Model Overview

- Utility for child i of attending or not school in time (age) t is:

$$u_{it}^s = \gamma\delta g_{it} + \mu_i + \alpha' z_{it} + \lambda ed_{it} + 1(p_{it} = 1)\beta^p x_{it}^p + 1(s_{it} = 1)\beta^s x_{it}^s + \epsilon_{it}$$
$$u_{it}^w = \delta w_{it}$$

- Unobserved heterogeneity:
  - $\Rightarrow$ ϵ_{it} is an iid logistic shock to costs of schooling
  - $\Rightarrow$ μ_i is permanent unobserved heterogeneity with discrete support

- Child may not be successful in completing a grade (exogenous prob. p_{it}^s, estimated from the data)

Value Functions

- Terminal value function (returns to schooling):

$$V(ed_{i,18}) = \frac{\alpha_1}{1+\exp(-\alpha_2 ed_{i,18})}$$

$\Rightarrow$ $ed_{i,18}$ acts as a sufficient statistics

- Value of school and of work at age t can be written as:

$$\begin{aligned} V_{it}^s(ed_{it}|\Omega_{it}) &= u_{it}^s + \beta\{p_{it}^s(ed_{it}+1)\mathbb{E}\max\left[V_{it+1}^s(ed_{it}+1), V_{it+1}^w(ed_{it}+1)\right]\} \\ &\quad + \{(1-p_{it}^s(ed_{it}+1))\mathbb{E}\max\left[V_{it+1}^s(ed_{it}), V_{it+1}^w(ed_{it})\right]\} \\ V_{it}^w(ed_{it}|\Omega_{it}) &= u_{it}^w + \beta\mathbb{E}\max\left[V_{it+1}^s(ed_{it}), V_{it+1}^w(ed_{it})\right] \end{aligned}$$

$\Rightarrow$ $\mathbb{E}\max$ functions have closed form expressions due to the logistic in $u_{it} = \tilde{u}_{it} + \epsilon_{it}$

Wages and General Equilibrium Responses

- Standard Mincer-type wage equation estimated outside of the schooling model:

$$\ln w_{ij} = q_j + a_1 age_i + a_2 educ_i + a_3 IMR_i + \omega_{ij}$$
$$q_j = b_1 \ln w_j^{ag} + b_2 P_j$$

⇒ Notice that the market wage does not depend on children's education

⇒ q_j represents the log price of human capital in the locality

⇒ $\ln w_j^{ag}$ is a sufficient statistic for the labor demand in the local area

⇒ The *Progresa* program $P_j \in \{0, 1\}$ pushes up child wages by decreasing child labor

⇒ No or limited selection on unobserved ability ($\hat{a}_3 < 0$ and very small)

Initial Conditions

- Past education decisions are unobserved and correlated with unobserved ability μ_i
- Ordered Probit model:

$$P(ed_{it} = e \mid z_{it}, x^s_{it}, x^p_{it}, h_i, w_{it}, \mu_i)$$

$\Rightarrow$ μ_i is added to the normally distributed random variable of the ordered probit,

$\Rightarrow$ h_i reflects past schooling costs, such as distance to closest school in the past

$\Rightarrow$ Past distance to schools should not affect current school participation decisions

Identification and Estimation

- Parameters of the (joint) model are estimated by simulated maximum likelihood (SMLE)

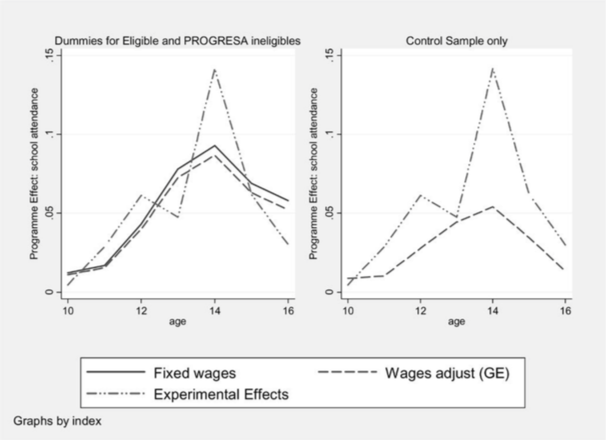
$$L_i = \sum_k P(at_i = 1 \mid z_{it}, x^s_{it}, x^p_{it}, ed_{it}, w_{it}, \mu_i) P(ed_{it} = e \mid z_{it}, x^s_{it}, x^p_{it}, h_i, w_{it}, \mu_i) P(\mu_i = \mu_i^k)$$

$\Rightarrow$ Schooling model using treatment-control variation + different ages in the same grade

$\Rightarrow$ Wage equation using treatment-control variation

$\Rightarrow$ Initial conditions model using exclusion restriction on past distance to school h_i

In-sample Fit of the Attanasio et al. Model



Principles for combining design-based and model-based analysis

Validation vs. Identification?

- Counterfactuals from Todd-Wolpin may be more credible
- But the framework in Attanasio-Meghir-Santiago is more parsimonious, and yet more general

$\Rightarrow$ Difficult to account for behavioral responses using only the control group

- One should use (at least some of the) policy variation for identification/estimation
- Then, if possible, use extra sources of variation for out-of-sample validation

The Data-then-Model Structure

- Show your data/variation with descriptive analysis
- Use the design-based analysis to provide preliminary evidence
- Clearly articulate the value-added of the model
- Use the design-based analysis to guide modeling choices and identification
- Choose parameters and counterfactuals that are linked to your variation

The Model-then-Data Structure

- Start with the model when the conceptual idea is the main contribution of the paper
- Present descriptive analysis that allows you to justify key model assumptions
- Use the design-based analysis as a source of variation for model estimation/validation
- Design counterfactuals that quantify the importance of the key channels in the model